

Kobo KEPUB — Gurmukhi Lagaan Clipping at Page Tops

Bug report · Japji Sahib e-ink EPUB project · May 2026

ASK

We are building a Gurmukhi EPUB for Kobo e-ink readers and have exhausted all known CSS and font-metric approaches to fix ascending vowel mark (lagaan) clipping at the top of pages in KEPUB. We are asking for Kobo's help on three specific points:

1. Which font metric does the KEPUB renderer use for the line box content area on the **first line of a new page** — `hhea.ascent` , `OS/2.sTypoAscender` , or something else?
2. Is there a supported CSS property or EPUB technique that reliably preserves space above the first line on a new page in KEPUB?
3. Is this a known difference between Kobo Clara Color and Kobo Libra Color, and if so, is a fix planned?

PROBLEM

In a Gurmukhi EPUB (Japji Sahib, TiroGurmukhi font, EPUB3 converted to KEPUB via kepubify), ascending vowel marks — sihari ੀ, bihari ੀ, dulavaa ੀ — are clipped at the top of every new page on **Kobo Clara Color**. The same marks render correctly on every line that is not the first line of a page.

Root cause: In KEPUB, the top half-leading is stripped from the first line on each new page. For Latin fonts this is invisible — Latin glyphs sit within the declared `hhea.ascent` . For Gurmukhi (and Indic scripts generally), vowel marks routinely exceed `hhea.ascent` and rely on that half-leading buffer. When it is stripped, they clip against the page boundary.

Key font metrics for TiroGurmukhi (embedded, OFL):

Metric	Value	Notes
<code>hhea.ascent</code>	755u → patched to 930u	Raised at build time to cover all lagaan
<code>OS/2.sTypoAscender</code>	755u → patched to 930u	USE_TYPO_METRICS flag is set; raised to match
<code>OS/2.usWinAscent</code>	1447u	Unchanged — already well above all glyph yMax values
Highest lagaan yMax	918u (dulavaa ੀ)	Below patched content area of 930u
Sihari / bihari yMax	884u	Both identical

Device behaviour:

- **Kobo Clara Color** — clipping on all pages, all font sizes, persists even with the Kobo reading header disabled
- **Kobo Libra Color** — no clipping after `border-top` fix (see Solutions); that renderer preserves `border-top` on page-first elements where Clara does not

This issue would affect any Indic script font where vowel marks exceed `hhea.ascent` , which is common — Devanagari, Bengali, Gurmukhi, Tamil and others all have ascending marks that extend above typical hhea values.



Kobo Clara Color — sihari and bihari clipped at top of page. Illustrative diagram using the embedded TiroGurmukhi font; the pink band marks the ~7px clipped by the renderer.



Kobo Libra Color — same EPUB, lagaan intact after `border-top` fix. The green bracket shows the 0.25 em gap that the renderer preserves above the first line.

SOLUTIONS TRIED

Approach	Outcome	Notes
<code>@page { margin-top: 48px }</code>	No fix	Moves the page boundary down; lagaan still bleed above it. Clara does not appear to honour <code>@page</code> for content layout in KEPUB.
<code>div { border-top: 0.25em solid transparent }</code> on each line wrapper	Fixed Libra / No fix Clara	Libra Color preserves <code>border-top</code> on page-first elements; Clara strips it along with <code>padding-top</code> and <code>margin-top</code> .
Patch embedded font: raise <code>hhea.ascent</code> and <code>OS/2.sTypoAscender</code> from 755u to 930u at build time	No fix	All lagaan (max 918u) now fall within the patched content area of 930u. Clipping continues on Clara, suggesting the renderer uses a different metric or a simplified em-box model that ignores these values entirely.
<code>white-space: nowrap !important</code> , <code>overflow-wrap: normal !important</code> on word spans	Separate issue	Addressed word-splitting at large font sizes (separate from clipping). Not related to the lagaan clipping problem.
Grapheme-cluster-level <code></code> wrapping in shaper	Separate issue	Addressed kepubify splitting combining marks from base consonants. Not related to the lagaan clipping problem.

The inconsistency between Clara Color and Libra Color — where the same EPUB and same CSS produce different results — strongly suggests a renderer version difference. We would like to understand what metric or model Clara is actually using so we can implement a correct workaround, or to know if this is a bug being tracked on Kobo's side.